VerifAI

Team Sleepyyy

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The Problem

KYC Verification at Scale is Inefficient and Unreliable

Current systems struggle with the sheer volume and complexity of document processing in India.

- India processes **millions of KYC verifications annually** across banking, insurance, and fintech sectors.
- Existing systems rely on **manual review and basic OCR-based tools**, which lack modern intelligence.

Key Limitations

No reasoning capability
Systems cannot interpret context, variations, or inconsistencies.

High false rejection rate (20–30%)
Valid applications are rejected due to rigid rule matching.

Lack of explainability
Decisions provide no audit trail or justification for outcomes.

Inadequate fraud detection
Tampered or edited documents often go undetected by basic tools.

The Solution: VerifAI

An AI-Powered KYC Verification Agent

VerifAI introduces an **agent-based approach** to document verification that goes beyond extraction and enables **reasoning, validation, and explainability**.

Structured Extraction

Uses Vision-Language Models to extract key fields with confidence scores

Tamper Detection

Detects edited documents using Error Level Analysis (ELA) and EXIF metadata

Cross-Verification

Performs semantic matching across documents to identify inconsistencies

Automated Compliance

Applies RBI KYC and AML rules through an auditable rule engine

Explainable Decisions

Generates clear, human-readable justifications for every outcome

Outcome: Reliable, Transparent, and Automated KYC Processing

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Identity Verification Pipeline

INPUT DOCUMENTS
Aadhaar | PAN | Utility Bill

PREPROCESSING
Base64 Encoding

UNIFIED PROMPT
GPT-4o Vision Processing

EXTRACTION
Structured JSON Output

MAPPING
LLM Reasoning & Validation

FUZZY MATCHING
Typo Resilience Scoring

CONTEXTUAL LOGIC
Consistency Checks

FINAL OUTPUT (DECISION)
✔ Match / ✖ Mismatch / ⚠️ Escalate
Confidence Score + Automated Narrative Explanation

Automated Forensic Workflow for Multi-Document Verification

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Modular Architecture for Absolute Integrity

STAGE 1

ELA

Detecting digital tampering.

STAGE 2

AI Multimodal Extraction

Unified vision processing.

STAGE 3

Identity Mapping

Resolving name variations & typos.

STAGE 4

Compliance Engine

Regional rule validation (Regex/Freshness).

STAGE 5

Risk Analysis

Synthetic detection & PEP/AML scanning.

STAGE 6

Executive Summary

Automated forensic narration.

Securing Verification with Intelligent Modular Stages

GPT-4o Cross-Document Intelligence

UNIFIED PROMPTING

Multi-Document Vision Context

Aadhaar, PAN, and Utility Bills processed simultaneously in a single vision context for holistic analysis.

BEYOND OCR

Cross-Document Mapping

The AI evaluates deep semantic relationships between different documents to detect structural inconsistencies.

TYPO RESILIENCE

Intelligent Fuzzy Matching

Sophisticated matching allows for realistic spelling variations while maintaining strict security and integrity.

CONTEXTUAL LOGIC

Real-Time Address Verification

Verifying addresses on utility bills against primary identity documents using real-time reasoning loops.

HOW WE OVERCAME THEM

SOLUTION 01

Document Variability

Used Vision-Language Models for flexible extraction

Schema-based output ensures consistency

SOLUTION 02

Fraud Detection

Implemented ELA + EXIF checks

Tampering detected before processing

SOLUTION 03

False Positives

Semantic matching (RapidFuzz)

Confidence thresholds + escalation

SOLUTION 04

Explainability

Multi-stage pipeline with reasoning trace

Human-readable justifications

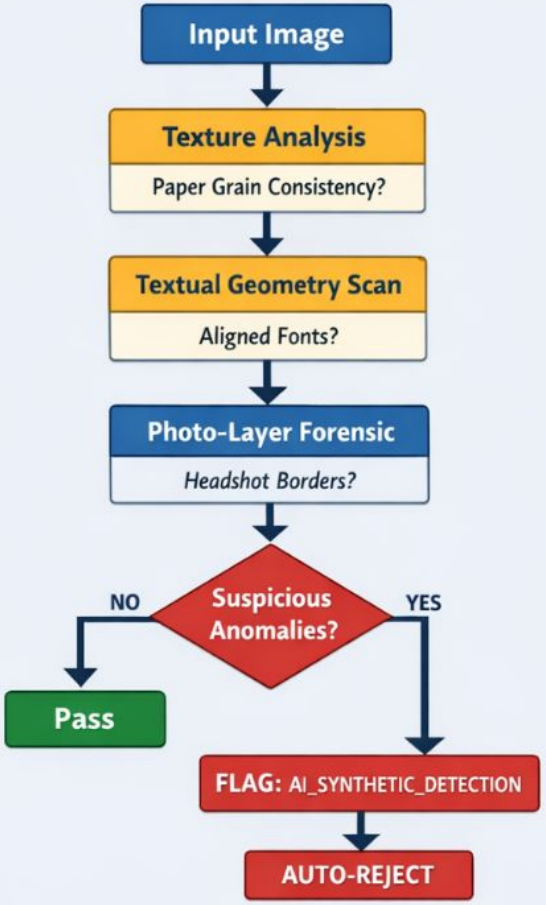
SOLUTION 05

System Scaling

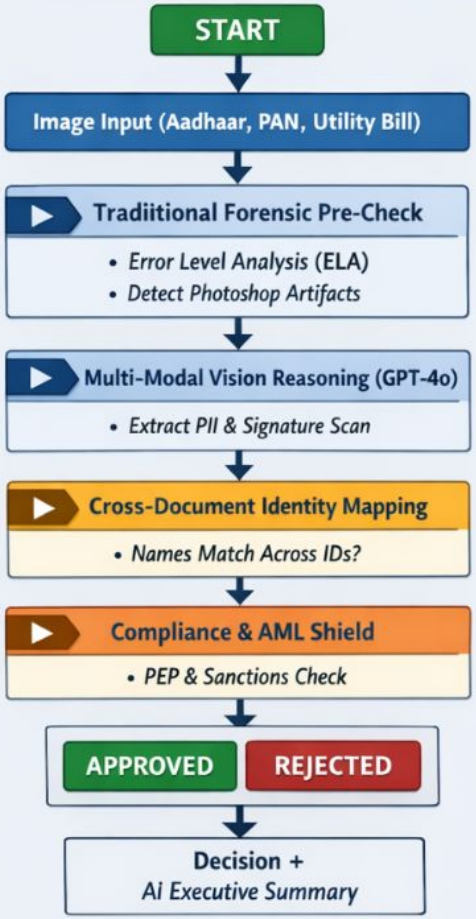
Event-driven (RabbitMQ/Redis)

Parallel efficient processing

Synthetic Detection Logic: *Deepfake Analysis*



VerifAI AI Engine: Forensic Brain: Identity Verification Process



Measuring the Success of VerifAI

01. ACCURACY & DECISION

Decision Quality

- False rejection rate **<10%**
- High AI vs ground truth alignment
- Approve / Escalate / Reject balance

02. FRAUD DETECTION

Performance Metrics

- Tampering detection (ELA accuracy)
- Stage 1 effectiveness
- Reduced fraud leakage

03. OPERATIONAL EFFICIENCY

Speed & Automation

- Verification **<10 seconds**
- 70–80% manual work reduction
- Faster onboarding pipeline

04. COMPLIANCE

Explainability

- 100% reasoning trace
- Clear rule-level justifications
- Audit-ready regulatory outputs

05. SYSTEM PERFORMANCE

Scalability

- Concurrent request handling
- Stable latency under load
- Fault-tolerant architecture

Success is building a system that is reliable, explainable, and scalable in real-world conditions.

Thank you